

Recursive Estimation	
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1	Probability Theory
1.1	Random Variables
1.1.1	Discrete Random Variables
DRV – Discrete Random Variable defined by $p_x, \mathcal{X}$	
$\mathcal{X} \subset \mathbb{Z}$ of all possible outcomes $p_x(\bar{x}) \geq 0$ and $\sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x}) = 1$ or $\sum_{\bar{x} \in \mathcal{X}} p_{x y}(\bar{x} \bar{y}) = 1$	
Margin. / Sum Rule	$p(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy}(\bar{x}, \bar{y})$
Cond. / Product Rule	$p_{x y}(\bar{x} \bar{y}) := \frac{p_{xy}(\bar{x}, \bar{y})}{p_y(\bar{y})}$
Total Prob. Theorem	$p_x(\bar{x}) := \sum_{\bar{y} \in \mathcal{Y}} p_{x y}(\bar{x} \bar{y})p_y(\bar{y})$
1.1.2	Continuous Random Variables
CRV – Continuous Random Variable: DRV $\sum \rightsquigarrow$ CRV $\int$	
- Set $\mathcal{X} \subset \mathbb{R}$ section of real line - PDF p_x satisfies $p_x(\bar{x}) \geq 0$ and $\int_{\mathcal{X}} p_x(\bar{x})d\bar{x} = 1$	
Caution: $Pr(x = \bar{x}) = 0, \quad Pr(x \in (a, b)) = \int_a^b p(\bar{x})d\bar{x}$	
1.1.3	Conditional PDF
Conditional PDF	
Margin. / Sum Rule	$p(\bar{x} \bar{z}) = \sum_{\bar{y} \in \mathcal{Y}} p_{xy z}(\bar{x}, \bar{y} \bar{z})$
Cond. / Product Rule	$p_{x yz}(\bar{x} \bar{y}, \bar{z}) := \frac{p_{xy z}(\bar{x}, \bar{y} \bar{z})}{p_{y z}(\bar{y} \bar{z})}$
Independence	
$p(x y) = p(x) \Leftrightarrow p(y x) = p(y) \Leftrightarrow p(x, y) = p(x)p(y)$	
$p(x, y, z) = p(x, y z)p(z) \rightarrow x, y$ indep. $p(x, y z) = p(z x, y)$	
Conditional Independence	
The knowledge of z makes x and y independent:	
$p(x y, z) = p(x z) \Leftrightarrow p(x, y z) = p(x z)p(y z)$	
Caution!!! in general we still have: $p(x, y) \neq p(x)p(y)$	
Caution!!!! Independence $\nRightarrow$ Conditional Independence	
1.2	Expectation and Variance
1.2.1	Expectation
Definition: Integral for CRV!	
$\mathbb{E}_x[x] = \sum_{\bar{x} \in \mathcal{X}} \bar{x}p_x(\bar{x})$	
Linearity	$\mathbb{E}_{xy}[a+bx+cy] = a+b\mathbb{E}_x[x]+c\mathbb{E}_y[y]$
Multi Variable	$\frac{\mathbb{E}_{xy}[g(x, y)]}{\sum_{\bar{y}} \sum_{\bar{x}} g(\bar{x}, \bar{y})p_{xy}(\bar{x}, \bar{y})} =$
Independence	$\mathbb{E}_{xy}[xy] = \mathbb{E}_x[x]\mathbb{E}_y[y]$
Law of Unconscious Statistician for $y = g(x)$	
$\mathbb{E}_y[y] = \sum_{\bar{y} \in \mathcal{Y}} \bar{y}p_y(\bar{y}) = \sum_{\bar{x} \in \mathcal{X}} g(\bar{x})p_x(\bar{x})$	
1.2.2	Variance (generally a matrix)
$\text{Var}_x[x] = \mathbb{E}_x[(x - \mathbb{E}_x[x])(x - \mathbb{E}_x[x])^\top] = \mathbb{E}_x[x^2] - \mathbb{E}_x[x]^2$	
Linearity	
$\text{Var}_x[a+bX+cY] = b^2 \text{Var}_X[x] + c^2 \text{Var}_Y[y] + 2bc\text{Cov}[X, Y]$	
Covariance: if independent $\text{Cov}(X, Y) = 0$	
$\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$	
1.3	Distribution Sampling
CDF – Cumulative Distribution Function	
$\hat{F}_x(\bar{x}) := \sum_{i=-\infty}^{\bar{x}} \hat{p}_x(i) = Pr(x \leq \bar{x}) = \int_{-\infty}^{\bar{x}} p_x(\hat{x})d\hat{x}$	
A1 – 1 DRV $\bar{u}$ sample of $u \sim \mathcal{U}$, solve for $\bar{x}$ such that:	
$\hat{F}_x(\bar{x} - 1) < \bar{u} \leq \hat{F}_x(\bar{x})$	
A2 – Multiple DRV or CRV	
Works for finite & infinite # of elements	

- Decompose $\hat{p}(\bar{x}, \bar{y}) = \hat{p}_{x y}(\bar{x} \bar{y})\hat{p}_y(\bar{y})$ - Apply A1 to get sample $\bar{y}$ via $\hat{p}_y(\bar{y})$ - With $\bar{y}$ fixed, apply A1 to get sample $\bar{x}$ via $\hat{p}_{x y}(\bar{x} \bar{y})$	A3 – 1 CRV $\hat{p}_x$: pw-conti., bounded. $u \sim \mathcal{U}(0, 1)$ and solve: $\bar{u} = F_x(\bar{x}) \rightarrow \bar{x} = F_x^{-1}(\bar{u})$
1.4 Change of Variables	
DRV: $x = g(y)$	
$p_x(\bar{x}) = \sum_{\bar{y} \in \mathcal{Y}: g(\bar{y}) = \bar{x}} p_y(\bar{y})$	
CRV general case $y = g(x), \rightarrow x = g^{-1}(y)$:	
$p_y(\bar{y}) = \frac{p_x(\bar{x})}{\left \frac{dg}{dx}(\bar{x})\right }, \quad \bar{x} = g^{-1}(y)$	
CRV – Multivar. & Indep. Case: $z = x + y$	
$p_z(\bar{z}) = \sum_{\bar{x} \in \mathcal{X}} p_x(\bar{x})p_y(\bar{z} - \bar{x})$	
CRV – Multivar. & Cond. Case: $z = g(w, x), w = h(z, x)$	
Assumptions: - $g(w, x), p_w$ cont. diff'bar	
- $g(w, x)$ strictly monotonic	
- For scalars $\det(x)$ is just x	
$p_{z x}(\bar{z} \bar{x}) = p_w(h(\bar{z}, \bar{x}) \bar{x}) \cdot \det(\frac{\partial}{\partial w}g(h(\bar{z}, \bar{x}), \bar{x})) ^{-1}$	
1.5 Miscellaneous	
Indep. multiplication: $z = xy \rightarrow p(z) = \sum_Y p_x(\frac{z}{y})p_y(y)$	
Uniform Distribution	
$p(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{else} \end{cases} \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \frac{a+b}{2} \\ \text{Var}\{x\} &= \frac{(b-a)^2}{12} \end{aligned} \right.$	
Triangular Distribution	
$p(x) = \begin{cases} \frac{2(x-a)}{(b-a)(c-a)} & a \leq x \leq c \\ \frac{2(b-x)}{(b-a)(b-c)} & c \leq x \leq b \\ 0 & \text{else} \end{cases}$	
$F(x) = \begin{cases} 0 & x < a \\ \frac{(x-a)^2}{(b-a)(c-a)} & a \leq x \leq c \\ 1 - \frac{(b-x)^2}{(b-a)(b-c)} & c \leq x \leq b \\ 1 & x > b \end{cases}$	
$\mathbb{E}\{x\} = \frac{a+b+c}{3}, \quad \text{Var}\{x\} = \frac{a^2+b^2+c^2-ab-ac-bc}{18}$	
Gaussian Distribution	
PDF Gaussian distributed vector CRV $y = (y_1, \dots, y_D)$	
$p(x) = \frac{1}{(2\pi)^{D/2} \det(\Sigma)^{1/2}} \exp\left(-\frac{(x - \mu)^T \Sigma^{-1} (x - \mu)}{2}\right)$	
$\rightsquigarrow$ fully characterized by mean $\mu \in \mathbb{R}^D$ and var $\Sigma \in \mathbb{R}^{D \times D}$	
(Scalar: $D = 1, \sigma^2 = \det(\Sigma) \rightarrow \det(\Sigma)^{1/2} = \sigma$)	
$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad \left \quad \begin{aligned} \mathbb{E}\{x\} &= \mu \\ \text{Var}\{x\} &= \sigma^2 \end{aligned} \right.$	
Sum of two independent Gaussians:	
$x_1 \sim \mathcal{N}(\mu_1, \Sigma_1), x_2 \sim \mathcal{N}(\mu_2, \Sigma_2) \rightarrow z = M_1x_1 + M_2x_2$	
$z \sim \mathcal{N}(M_1\mu_1 + M_2\mu_2, M_1\Sigma_1M_1^T + M_2\Sigma_2M_2^T)$	
Diagonal Variance with $\Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_D^2), \mu = [\mu_D^1]$	
$\rightsquigarrow y_i$ mut.indep. iff Σ diag.	
$p(y) = \prod_{i=1}^D \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp\left(-\frac{(y_i - \mu_i)^2}{2\sigma_i^2}\right)$	
Dependent Gaussians:	
$\Sigma_{x \pm y} = \Sigma_x + \Sigma_y \pm \Sigma_{xy} \pm \Sigma_{xy}^T, \mu_{x \pm y} = \mu_x \pm \mu_y$	

2	Bayesian Tracking
3	Extracting Estimates
4	Kalman Filter
5	Particle Filter
6	Observer Based Control